



The Ground Plane (x, z)

Stop looking at the wall — look down

벽에서 바닥으로 — 이제 (x, z) 를 써요

BIG PICTURE

The floor uses (x, z) 



x is across. z is forward. y is still height.

바닥에서는 가로 x, 앞으로 z

HOW TO COUNT

Count the axis — one block, one number



1 Stand on the start block — that spot is $(0, 0, 0)$.

2 $x \rightarrow$ walk **right**. Add **+1** for every block.

3 $y \rightarrow$ jump **up**. Add **+1** for every block.

4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

Wall vs floor



wall (x, y)

across + UP
(standing up)



floor (x, z)

across + FORWARD
(lying flat)

벽은 위로(y), 바닥은 앞으로(z)

CONCEPT

The three directions

x
across →

y
up ↑

z
forward ↗

On the flat floor we keep $y = 0$ and move with x and z .

바닥은 y 를 0으로 두고 x , z 만 움직여요

CODE

A path running forward (grow z)

```
for z in range(5):  
    blocks.place(STONE, pos(2, 0, z))
```

x stays 2, y stays 0 (on the ground), z grows $0 \rightarrow 4$. A path going **forward**.

z가 자라면 앞으로 길이 생겨요

CODE

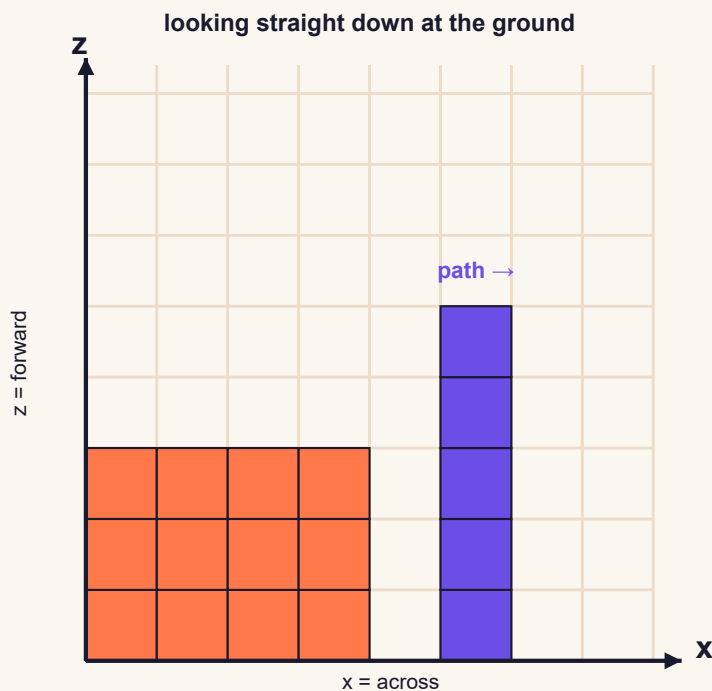
Filling a whole floor

```
for z in range(3):          # 3 forward
    for x in range(4):      # 4 across
        blocks.place(STONE, pos(x, 0, z))
```

Inner loop runs x across, outer steps z forward: a floor pad **4 × 3**.

SEE IT

The floor from above (x, z)



바닥을 위에서 본 모습 — x는 가로, z는 앞으로(위쪽)

PREDICT 🤔

What shape covers the floor?

Predict first — then click reveal

```
z = 0
repeat 3 times:
  x = 0
  repeat 4 times:
    place stone block at (x, z)
    add 1 to x
  add 1 to z
```

Reveal output 🔍 reveal

VOCAB

New words today

floor / ground

the flat plane you look down on

바닥

z

how far forward (depth)

깊이 / 앞으로

forward

the direction z grows

앞으로

path

a line of floor blocks to walk on

길



What's wrong with this code?

Goal: a path running **forward** at $x = 2$.

buggy

```
n = 0
repeat 5 times:
    place stone block at (n, 2)
    add 1 to n
```

On the floor, the slots are (x, z) . Think first.



DEBUG

The bug

Bug: **n** is in the **x** slot, so the path runs across instead of forward. Put the growing number in the **z** slot and keep **x** = 2.

fixed

```
z = 0
repeat 5 times:
  place stone block at (2, z)
  add 1 to z
```

YOUR TURN!

Lay a floor + path

- ✓ Open your homework world + Code Builder
- ✓ Fill a floor pad with a nested loop (x across, z forward)
- ✓ Lay a path that runs forward (grow z) or across (grow x)
- ✓ Place a marker at `(0, 0, 0)` where the floor starts
- ✓ Send a photo looking down at your floor



QUESTION 1 OF 3

Which two numbers describe a spot on the **floor**?

Just x

x and z

x and y

y and z



QUIZ

QUESTION 2 OF 3

What does growing **z** do on the floor?

Moves across.

Nothing.

Moves up into the air.

Moves forward.



QUIZ

QUESTION 3 OF 3

Your floor comes out as a thin line, not a rectangle. What's missing?

A bigger inner range.

Growing z in the outer loop.

Growing y in the outer loop.

Nothing.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Floor laid!

Next: layer up — terrain & trees.

다음 주: 위로 쌓아 지형과 나무 만들기